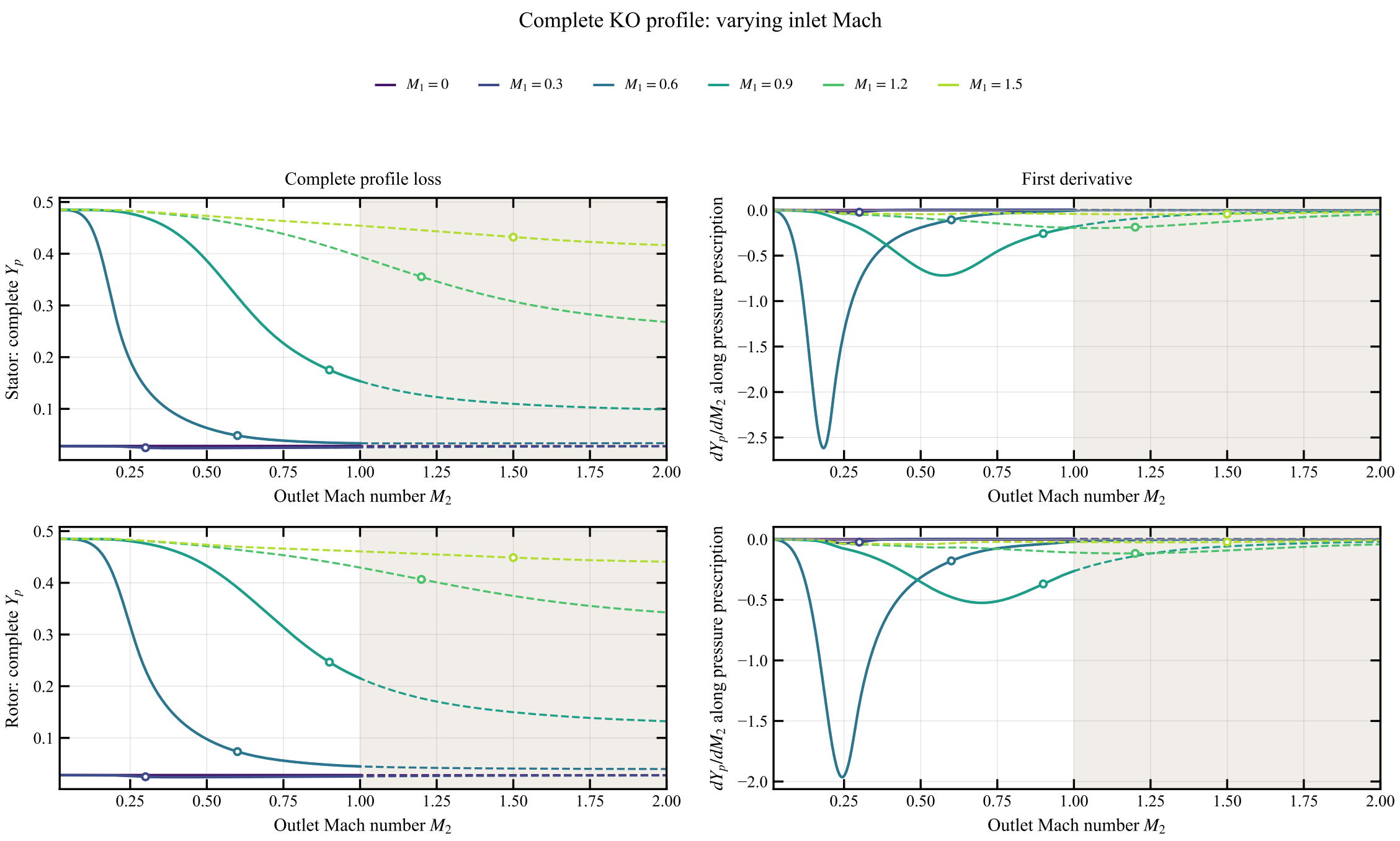
Complete KO profile: varying inlet Mach
$M_1 = 0$
$M_1 = 0.3$
$M_1 = 0.6$
$M_1 = 0.9$
$M_1 = 1.2$
$M_1 = 1.5$
Complete profile loss
First derivative
Stator: complete Y_p
Rotor: complete Y_p
dY_p/dM_2 along pressure prescription
Outlet Mach number M_2
Angle = 60 deg; s/c = 0.7; z = 0.5; thickness/chord = 0.2; Re = 500000; hub/tip = 0.8.
Prescribed gamma = 1.4, equal row-frame p0 = 200 kPa; static pressures vary with Mach.
All M1 > M2 cases included; circles: M1 = M2. Tint: M2 > 1. Dashed: M2 > 1, hub M1 > 1.2 or hub/tip < 0.5.